

CENTRAL BOARD OF SECONDARY EDUCATION**HALF-YEARLY PRACTICE EXAMINATION (SESSION 2024-2025)****SUBJECT: SCIENCE (CLASS - IX) | CHAPTER 6: HOW FORCES AFFECT MOTION [SET C - NCERT EXEMPLAR SPECIAL]****TIME ALLOWED: 45 MINUTES****MAXIMUM MARKS: 25****PAPER CODE: 086/CH06/SET-C****Student's Name:** _____**Roll No.:** _____**Class & Section:** Class IX - _____**Date of Exam:** _____**Invigilator's Sign:** _____**GENERAL INSTRUCTIONS (SET C - NCERT EXEMPLAR & CRITICAL THINKING):**

1. The question paper comprises **25 Multiple Choice Questions (MCQs)** of **1 mark each**.
2. **Section A (Q1 – Q15):** NCERT Exemplar based Multiple Choice Questions with single correct option.
3. **Section B (Q16 – Q20):** Deep Assertion-Reasoning based questions testing scientific cause-and-effect.
4. **Section C (Q21 – Q25):** Case-Study / Laboratory Experimental Investigation questions.
5. All questions are compulsory. There is no negative marking.
6. Darken the corresponding circle on the **OMR Sheet** completely using a blue/black ballpoint pen.

SECTION A: NCERT EXEMPLAR MCQS (Q.1 TO Q.15)**[15 MARKS]**

Q.1 In an ideal Atwood machine, two masses $m_1 = 3\text{ kg}$ and $m_2 = 2\text{ kg}$ are connected by a light string over a frictionless pulley. What is the acceleration of the system ($g = 10\text{ m/s}^2$)?

- (A) 1 m/s^2 (B) 2 m/s^2
(C) 5 m/s^2 (D) 10 m/s^2

Q.2 What is the tension force in the string of the above Atwood machine?

- (A) 20 N (B) 24 N
(C) 30 N (D) 50 N

Q.3 When a person walks forward on horizontal ground, what is the direction of the friction force exerted by the ground on the sole of their foot?

- (A) In the backward direction (B) In the forward direction
(C) Vertically downward (D) Zero

Q.4 Sand is dropped onto a conveyor belt moving horizontally at a constant speed of 2 m/s at a rate of 5 kg/s . What extra force must the motor exert to keep the belt moving at constant speed?

- (A) 2.5 N (B) 10 N
(C) 20 N (D) 5 N

Q.5 A force-time graph for an impact shows a triangle of base $\Delta t = 0.04\text{ s}$ and peak force $F_{\text{max}} = 1000\text{ N}$. What is the total impulse delivered?

- (A) 40 N·s (B) 20 N·s
(C) 80 N·s (D) 10 N·s

Q.6 A stationary bomb of mass 9 kg explodes into two fragments of masses 3 kg and 6 kg. If the kinetic energy of the 3 kg fragment is 216 J, what is the kinetic energy of the 6 kg fragment?

- (A) 432 J (B) 108 J
(C) 216 J (D) 72 J

Q.7 A body of mass 2 kg is acted upon by two mutually perpendicular forces of 6 N and 8 N. What is the magnitude of the resulting acceleration?

- (A) 5 m/s^2 (B) 7 m/s^2
(C) 10 m/s^2 (D) 14 m/s^2

Q.8 A 5 kg block rests on an inclined plane making an angle of 30° with the horizontal. What is the component of gravity acting parallel down the incline ($g = 10 \text{ m/s}^2$)?

- (A) 50 N (B) 25 N
(C) 43.3 N (D) 10 N

Q.9 Why does a heavy wooden box at rest require more force to start moving than to keep it sliding at constant speed?

- (A) Rolling friction is higher than static friction (B) Limiting static friction is greater than kinetic (sliding) friction
(C) Mass decreases once sliding begins (D) Gravity drops while in motion

Q.10 A bullet of mass 10 g leaves a gun barrel of length 0.8 m with a muzzle velocity of 400 m/s. Assuming uniform acceleration, the average force exerted on the bullet inside the barrel is:

- (A) 1000 N (B) 2000 N
(C) 500 N (D) 100 N

Q.11 Which law of Newton provides the conceptual definition of force as an external agency that alters state of motion?

- (A) First law of motion (B) Second law of motion
(C) Third law of motion (D) Law of gravitation

Q.12 An object of mass m is suspended by a string from the ceiling of a car accelerating forward with acceleration a . The string tilts backward at an angle θ given by:

- (A) $\tan \theta = a/g$ (B) $\tan \theta = g/a$
(C) $\sin \theta = a/g$ (D) $\cos \theta = a/g$

Q.13 A hovercraft gliding over a frozen lake with its engine turned off continues moving for kilometers because:

- (A) It generates infinite thrust (B) Air cushion eliminates virtually all contact friction, illustrating Galileo's law of inertia
(C) Lake ice rotates (D) Gravity pulls it forward

Q.14 Two skaters on smooth ice face each other. Skater A (60 kg) pushes Skater B (40 kg) with a force of 120 N. What force does Skater B exert on Skater A?

- (A) 80 N (B) 120 N
(C) 180 N (D) Zero

Q.15 What is the ratio of accelerations of Skater A (60 kg) to Skater B (40 kg) from the push above?

- (A) 2 : 3 (B) 3 : 2
(C) 1 : 1 (D) 4 : 9

SECTION B: ASSERTION - REASON QUESTIONS (Q.16 TO Q.20)

[5 MARKS]

Directions for Q16 to Q20: Select the correct option from the choices given below:

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true (or both are false).

Q.16 Assertion (A): Linear momentum of an isolated system of interacting particles is conserved.

Reason (R): Internal forces between particles in an isolated system always cancel out in action-reaction pairs.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.17 Assertion (A): When an apple falls towards the Earth, the Earth also accelerates towards the apple.

Reason (R): The gravitational force on the Earth by the apple is equal to the gravitational force on the apple by the Earth, but the Earth's enormous mass makes its acceleration undetectable.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.18 Assertion (A): A cyclist leans inward towards the center while negotiating a curve on a level road.

Reason (R): Leaning creates a torque from the normal contact force and gravity that balances the overturning torque produced by centrifugal effect.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.19 Assertion (A): Mass is an invariant scalar measure of the quantity of matter in a body.

Reason (R): The weight of an object remains identical whether measured on Earth, on the Moon, or in deep outer space.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.20 Assertion (A): Seatbelts in automobiles prevent injury by applying an external backward force on occupants during sudden deceleration.

Reason (R): Occupants tend to maintain their forward velocity due to inertia of motion.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

SECTION C: EXPERIMENTAL INVESTIGATION & CASE STUDY (Q.21 TO Q.25)

[5 MARKS]

READING PASSAGE & EXPERIMENTAL INVESTIGATION:

Case Study: Linear Air Track Elastic Collision Experiment

Two gliders, Glider 1 ($m_1 = 0.40 \text{ kg}$) and Glider 2 ($m_2 = 0.20 \text{ kg}$), are placed on a linear air track where cushions of compressed air eliminate friction. Glider 1 is pushed towards stationary Glider 2 ($u_2 = 0$) with an initial velocity $u_1 = 3.0 \text{ m/s}$. The gliders are equipped with spring bumpers so that the collision is completely elastic. After collision, Glider 1 continues forward at $v_1 = 1.0 \text{ m/s}$ while Glider 2 shoots forward at velocity v_2 .

Q.21 What is the initial total momentum of the two-glider system before collision?

- (A) $0.60 \text{ kg}\cdot\text{m/s}$
 (B) $1.20 \text{ kg}\cdot\text{m/s}$
 (C) $1.80 \text{ kg}\cdot\text{m/s}$
 (D) $2.40 \text{ kg}\cdot\text{m/s}$

Q.22 Using the law of conservation of momentum, what is the final velocity v_2 of Glider 2 after impact?

- (A) 2.0 m/s
 (B) 4.0 m/s
 (C) 3.0 m/s
 (D) 5.0 m/s

Q.23 What is the total kinetic energy of the system before collision?

- (A) 0.90 J (B) 1.80 J
(C) 3.60 J (D) 1.20 J

Q.24 What is the total kinetic energy of the system after collision?

- (A) 1.80 J (B) 1.60 J
(C) 2.00 J (D) 1.20 J

Q.25 If the spring bumpers were replaced with sticky Velcro pads so the gliders stuck together upon collision, the collision would be classified as:

- (A) Super-elastic (B) Perfectly inelastic
(C) Partially elastic (D) Conservative

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET C)

01	(A) (B) (C) (D)	02	(A) (B) (C) (D)	03	(A) (B) (C) (D)	04	(A) (B) (C) (D)	05	(A) (B) (C) (D)
06	(A) (B) (C) (D)	07	(A) (B) (C) (D)	08	(A) (B) (C) (D)	09	(A) (B) (C) (D)	10	(A) (B) (C) (D)
11	(A) (B) (C) (D)	12	(A) (B) (C) (D)	13	(A) (B) (C) (D)	14	(A) (B) (C) (D)	15	(A) (B) (C) (D)
16	(A) (B) (C) (D)	17	(A) (B) (C) (D)	18	(A) (B) (C) (D)	19	(A) (B) (C) (D)	20	(A) (B) (C) (D)
21	(A) (B) (C) (D)	22	(A) (B) (C) (D)	23	(A) (B) (C) (D)	24	(A) (B) (C) (D)	25	(A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & EXEMPLAR EXPLANATIONS (SET C)

Q. No.	Correct	Scientific Rationale & NCERT Exemplar Analysis
Q.1	(B)	$a = \frac{m_1 - m_2}{m_1 + m_2} g = \frac{3 - 2}{3 + 2} \times 10 = \frac{1}{5} \times 10 = 2 \text{ m/s}^2$.
Q.2	(B)	$T = \frac{2 m_1 m_2}{m_1 + m_2} g = \frac{2(3)(2)}{3 + 2} \times 10 = \frac{12}{5} \times 10 = 24 \text{ N}$.
Q.3	(B)	To walk forward, the foot pushes backward against the ground. By Newton's third law, the ground exerts a static friction force in the forward direction on the foot, propelling the person.
Q.4	(B)	$F = v \frac{dm}{dt} = 2 \text{ m/s} \times 5 \text{ kg/s} = 10 \text{ N}$.
Q.5	(B)	$\text{Impulse} = \text{Area under } F-t \text{ graph} = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 0.04 \times 1000 = 20 \text{ N} \cdot \text{s}$.
Q.6	(B)	By conservation of momentum, $ p_1 = p_2 = p$. Since $E_k = \frac{p^2}{2m}$, $E_k \propto 1/m$. $\frac{E_2}{E_1} = \frac{m_1}{m_2} = \frac{3}{6} = 0.5$. $E_2 = 216 \times 0.5 = 108 \text{ J}$.
Q.7	(A)	$F_{\text{net}} = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10 \text{ N}$. Acceleration $a = F_{\text{net}} / m = 10 / 2 = 5 \text{ m/s}^2$.
Q.8	(B)	$F_{\text{parallel}} = mg \sin 30^\circ = 5 \times 10 \times 0.5 = 25 \text{ N}$.
Q.9	(B)	At rest, microscopic surface irregularities interlock tightly (cold welding). Once sliding begins, asperities do not have time to interlock as deeply, so kinetic friction is less than limiting static friction.
Q.10	(A)	$v^2 = 2as \Rightarrow 400^2 = 2a(0.8) \Rightarrow 160,000 = 1.6a \Rightarrow a = 100,000 \text{ m/s}^2$. $F = ma = 0.01 \times 100,000 = 1000 \text{ N}$.
Q.11	(A)	Newton's first law qualitatively defines force; Newton's second law quantitatively measures force ($F = ma$).
Q.12	(A)	In the accelerating frame, pseudo force ma acts backward and gravity mg acts downward. $T \sin \theta = ma$, $T \cos \theta = mg \Rightarrow \tan \theta = a/g$.
Q.13	(B)	In the near absence of resistive frictional forces, an object in motion maintains its velocity indefinitely according to inertia.
Q.14	(B)	Newton's third law: action and reaction forces are strictly equal in magnitude and opposite in direction (120 N).
Q.15	(A)	$a_A = 120/60 = 2 \text{ m/s}^2$. $a_B = 120/40 = 3 \text{ m/s}^2$. Ratio $a_A : a_B = 2 : 3$.
Q.16	(A)	Since all internal interactions are equal and opposite, net internal force is zero. In the absence of external forces, $\frac{dP_{\text{total}}}{dt} = 0$, conserving momentum.
Q.17	(A)	$F = ma$. The force is equal, but since $M_{\text{Earth}} \approx 6 \times 10^{24} \text{ kg}$, $a_{\text{Earth}} = F / M_{\text{Earth}} \approx 10^{-24} \text{ m/s}^2$, which is imperceptible.
Q.18	(A)	Leaning inwards ensures the resultant ground reaction force passes through the center of mass, preventing the bike from toppling outward.
Q.19	(C)	Assertion is true. Reason is false: Weight is gravitational force ($W = mg$), which varies depending on local gravitational field ($g_{\text{Moon}} \approx g_{\text{Earth}}/6$, zero in deep space).
Q.20	(A)	Inertia carries unrestrained bodies forward; seatbelts provide the necessary decelerating force over a controlled distance.
Q.21	(B)	$P_{\text{initial}} = m_1 u_1 + m_2 u_2 = (0.40 \times 3.0) + (0.20 \times 0) = 1.20 \text{ kg} \cdot \text{m/s}$.
Q.22	(B)	$1.20 = m_1 v_1 + m_2 v_2 \Rightarrow 1.20 = (0.40 \times 1.0) + 0.20 v_2 \Rightarrow 1.20 = 0.40 + 0.20 v_2 \Rightarrow 0.20 v_2 = 0.80 \Rightarrow v_2 = 4.0 \text{ m/s}$.
Q.23	(B)	$E_{k, \text{initial}} = \frac{1}{2} m_1 u_1^2 = \frac{1}{2} (0.40)(3.0^2) = 0.20 \times 9 = 1.80 \text{ J}$.

Q. No.	Correct	Scientific Rationale & NCERT Exemplar Analysis
Q.24	(A)	$E_{k,final} = \frac{1}{2}(0.40)(1.0^2) + \frac{1}{2}(0.20)(4.0^2) = 0.20 + 0.10(16) = 0.20 + 1.60 = 1.80 \text{ J}$ (conserved in elastic collision).
Q.25	(B)	A collision where the bodies stick together and move with a common velocity experiences maximum kinetic energy loss, defined as perfectly inelastic.